INCH-POUND

MIL-R-39017/2J
19 March 1991
SUPERSEDING
MIL-R-39017/2H
3 July 1985

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, (INSULATED), ESTABLISHED RELIABILITY STYLE RLR20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RLR20, established reliability, insulated, film, fixed resistors.

2. APPLICABLE DOCUMENT

2.1 Government document.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-39017 - Resistors, Fixed, Film, (Insulated), Established Reliability, General Specification for.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

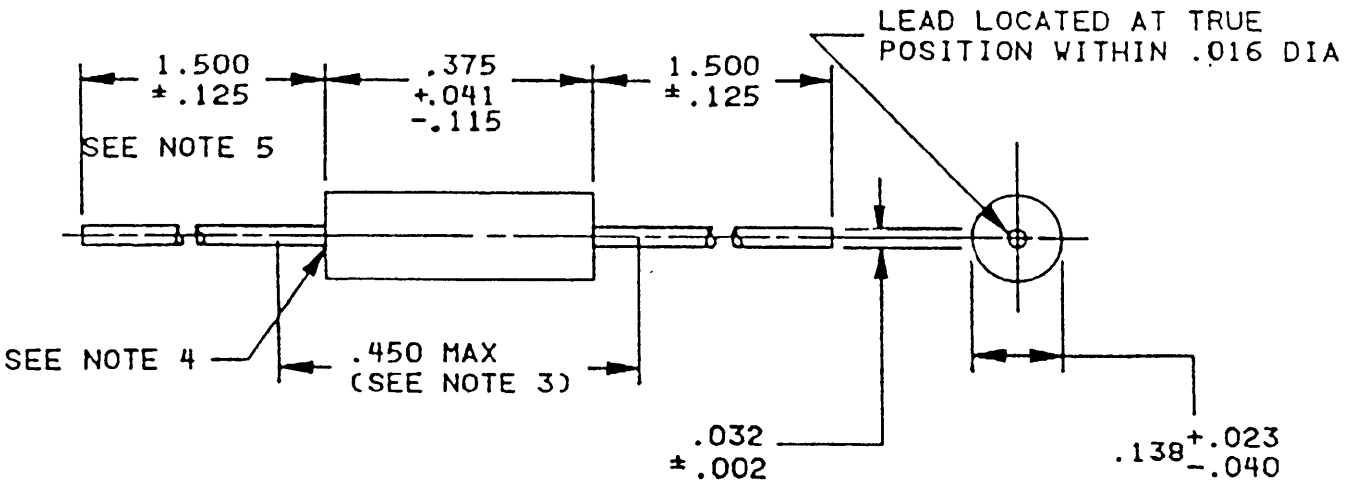
3.1 Requirements. Requirements shall be in accordance with MIL-R-39017, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Inches	mm
.002	0.05
.016	0.41
.023	0.58
.032	0.81
.041	1.04
.115	2.92
.125	3.18
.138	3.51
.375	9.53
.450	11.43
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Maximum length is "clean lead" to "clean lead".
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.
5. Lead length for tape and reel packaging shall be 1 inch minimum.

FIGURE 1. Style RLR20 resistor.

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3.3 Power rating. The power rating shall be 0.50 watt.

3.4 Voltage rating. The continuous working voltage shall not exceed 350 volts.

* 3.5 Resistance. Minimum and maximum resistance values for temperature characteristics of ±100 PPM and ±350 PPM shall be as follows:

	100 PPM	350 PPM
Minimum resistance	4.3Ω	3.3 MΩ
Maximum resistance	3.01 MΩ	22 MΩ

3.6 Maximum weight. The maximum weight shall not exceed .75 gram.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39017.

4.2 Power conditioning. The maximum voltages applied shall not exceed 350 volts, ac or dc.

4.3 Dielectric withstanding voltage. The test voltages applicable to style RLR20 are as follows:

Atmospheric pressure - 500 volts rms.

Barometric pressure - 250 volts rms.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39017.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-39017 are applicable to this specification.

6.2 Substitution data.

6.2.1 MIL-R-22684. Resistors of this specification, regardless of their failure rate designation, are substitutes for resistors of the same resistance value and tolerance specified in the inactivated specification sheet MIL-R-22684/2.

6.2.2 MIL-R-11. Resistors of this specification of resistance tolerance "G" are substitutes for style RC22 of MIL-R-11/4 in the following resistance range: 4.3 ohms through 1.0 megohm inclusive. Use MIL-R-39008/2, style RCR20, resistance tolerance "G" for RC22 values from 2.7 ohms through 3.9 ohms and 1.1 megohm through 22 megohms inclusive.

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-39017/2J	2. DOCUMENT DATE (YYMMDD) 19 March 1991

3. DOCUMENT TITLE
Resistors. Fixed, Film, (Insulated), Established Reliability Style RLR20

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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